

Personal Statement

I am applying for the PhD program in Physics at National University of Singapore to pursue my interest in exploiting first-principles modeling methods such as density functional theory (DFT), combined with data-driven techniques like machine learning (ML) to help understand, discover, and design material systems. I believe my research experience and training will enable me to excel in the PhD program.

Research Experience in first-principles studies My journey into first-principles studies began during my sophomore year, when the COVID-19 pandemic kept us at home and prevented access to physical laboratories. Fortunately, Prof. Zhang Peng offered me the opportunity to engage in DFT studies on the vibrational properties of ice phases. The DFT approach enabled effective computational experiments with sufficient accuracy. Building on this advantage, I transformed an old desktop computer into a computing workstation and started working on the Phonon Density of States (PDOS) for ice III. Over two months, I explored various modeling methods and computational parameters, ultimately achieving agreement between calculated PDOS and experimental data. As the far-infrared spectrum holds great potential due to emerging terahertz technology, my next goal is to explain the origin of this spectral range. I partnered with a graduate student to integrate DFT-simulated vibration normal modes with lattice dynamic analysis. Through a comparative analysis with ice IX, we proposed that the overlapping of two intrinsic hydrogen bond vibration modes, namely two-bond modes and four-bond modes, has led to the flat translational band in ice III. Furthermore, we attributed the splitting of degenerate vibrational modes to the hydrogen-disordered structure. This work led to my first published paper.

Upon graduating from Shandong University, our dean told us that "academic research is not exclusive to a few brilliant minds; anyone who dedicates 10 years to a field can achieve success" Encouraged by this advice, I decided to put it into practice. During my master program at NUS, I continued to work on first-principles studies under the supervision of *Prof. Quek Su Ying*. My master project aims to find promising quantum point defect in 2D WS₂. Specifically, the project focused on 4d and 5d transition metal (TM) substitutional impurities in multiple charge states. I developed a VASP screening workflow to identified deep-level defects with spin-triplet ground state. Performing large-scale calculations on the NUS HPC cluster has allowed me to use and hone skills in automation (Bash and Python), high-throughput processing (VASPKIT), job scheduling (PBS), and parallelization. As a result, three candidates were chosen for future evaluation on their magneto-optical properties. Additionally, I presented a statistical overview on the point groups for relaxed defects and their formation energies for all TM dopants. Given the limited research on WS₂, I also reviewed and drew insights from well-established studies on other material platforms for quantum defects, such as diamond and hBN, as well as the associated first-principles methodologies. These efforts culminated in my master thesis, including a comprehensive literature review and a detailed technical report.

Training in machine learning While I possess the passion for physical and material science, I also maintain substantive interests in data science and artificial intelligence. I have pursued a minor degree in artificial intelligence and actively engaged in data-driven and AI projects. It is exciting to find that AI is being extensively used to empower and accelerate material discovery.

In the AI minor program at Shandong University, I learned about the algorithms and techniques that underlie AI capabilities, including machine learning, natural language processing, computer vision, and robotics. The substantial coursework made it demanding to balance with my major degree studies, as I

often had to handle multiple projects on diverse topics simultaneously. Dealing with these challenges enabled me to cultivate useful skills in project and time management. My expertise in machine learning progressed further during my master's studies. I took *PC5252 Bayesian Statistics and Machine Learning*, where I acquired theoretical foundations from the Probabilistic Perspective. During *PC5251 Applied Machine Learning*, I delved into machine learning case studies within the physical science. Beyond my coursework, I spent three months learning and applying ensemble learning model to real-world datasets. During this time, I successfully reproduced highly-rated public solutions and taught myself advanced ensemble techniques, as well as developing skills in feature engineering and hyperparameter tuning.

Practice in numerical modelling My passion for applying computational methods to scientific challenges led me to explore various numerical modeling method. Motivated by multiscale modeling across different length scales, I chose finite element analysis (FEM) in Permanent Magnet Motors for my undergraduate capstone project in minor degree, where I gained basic knowledge in FEM analysis within Ansys Workbench. At NUS, inspired by *Prof. Wang Jian-sheng's* detailed lectures on Monte Carlo (MC) methods, I used the MC methods and convolutional neural networks to identify phase transitions in 2D Ising model. In my work with Bayesian neural networks, I employed stochastic gradient Langevin dynamics in MCMC sampling as an optimization method for model parameters.

Research Interest My specific research interest is in using multiscale approaches combined with machine learning to discover and understand materials. I am particularly interested in three areas of this research: Firstly, I plan to leverage machine learning to build efficient computational screening workflows, transforming the exhaustive search of all combinations in traditional routines. This includes utilizing transfer learning to take advantage of published databases and pre-trained models, and employing active learning to strategically explore potential chemical spaces, creating a modeling-search-computation iteration. Secondly, I am interested in applying interpretable machine learning techniques in material science to extract physically meaningful insights. This involves both intrinsically interpretable models, such as SISSO and LASSO, and extrinsic explanation methods, like SHAP. Thirdly, I aim to utilize multiscale modeling approaches to capture the multiphysical mechanisms that govern material behaviors. This includes integrating microscale (DFT and molecular dynamics), mesoscale (phase-field method and force-field modeling), and macroscale (Finite Element Methods) approaches.

I believe NUS would cater well to my aspirations because of the diverse strength of its faculty and students. In the field of computational material science, I would be excited to work with XXX, XXX and XXX. Their professional expertise and research pursuits are what I have been eagerly anticipating. After completing my PhD, I would like to be active in material modelling community, whether it is in academia or industry. With the rapid development of HPC technology and AI capabilities, computer simulations will affect how materials science is performed. I believe that NUS physics department would be the ideal environment for me to continue my studies and become a valuable and contributing member.